Lesson 7

Material

https://github.com/unixfile/learning-python-2026

Today

- Objects and classes
- Initialisation and `self`
- Class attributes vs instance attributes
- Instance and static methods
- Magic methods
- Protected and private attributes
- Properties
- Data classes
- Inheritance
- Composition

1 Objects and classes

- An object bundles data (**attributes**) and behaviour (**methods**)
- Everything in Python is an object: integers, strings, lists, functions
- Every object is created from a blueprint called a **class**

```
type(42)        # <class 'int'>
type('hello')   # <class 'str'>
type([1, 2])    # <class 'list'>
```

2 The `class` keyword

```
class Dog:
    species = 'Canis familiaris'   # class attr
    def __init__(self, name, age):
        self.name = name    # instance attr
        self.age  = age     # instance attr
    def bark(self):
        print(f'{self.name} says woof!')

rex   = Dog('Rex', 5)
pluto = Dog('Pluto', 3)

rex.bark()      # Rex says woof!
rex.name        # 'Rex'
Dog.species     # 'Canis familiaris'
```

- `Dog(...)` **instantiates** the class, producing a new object
- Each instance gets its own `name` and `age`

3 `self`: the hidden first argument

```
rex = Dog('Rex', 5)

rex.bark()          # what you write
Dog.bark(rex)       # what Python actually runs
```

- `rex.bark()` is shorthand: Python passes `rex` in as `self`
- So `self` is just the instance the method was called on
- You declare `self`, but you never pass it yourself
- `__init__` works the same way and runs automatically: `Dog('Rex', 5)` calls `__init__(rex, 'Rex', 5)`

4 Class attributes vs instance attributes

```
class Dog:
    species = 'Canis familiaris'    # shared
    def __init__(self, name):
        self.name = name    # per-instance

rex  = Dog('Rex')
lola = Dog('Lola')

Dog.species     # 'Canis familiaris'
rex.species     # inherited from the class
rex.name        # 'Rex'
```

- Class attributes live on the class and are shared
- Instance attributes live on the instance and are independent

5 Instance methods

- Methods defined inside a class that operate on an instance
- `self` is always the first parameter

```
class Dog:
    def __init__(self, name, age):
        self.name = name
        self.age  = age

    def bark(self):
        print(f'{self.name} says woof!')

    def dog_years(self):
        return self.age * 7

buddy = Dog('Buddy', 3)
buddy.bark()          # Buddy says woof!
buddy.dog_years()     # 21
```

6 Static methods

- Methods that belong to the class but do not receive `self`
- Useful for utilities that are logically related to the class

```
class MathOps:
    @staticmethod
    def add(x, y):
        return x + y

    @staticmethod
    def square(x):
        return x * x

MathOps.add(5, 3)    # 8
MathOps.square(4)    # 16

ops = MathOps()
ops.add(2, 2)    # 4 (works on an instance too)
```

7 Magic methods

Dunder methods let objects hook into built-in syntax:

```
__init__    called on creation
__str__     str(obj), print(obj)
__repr__    repr(obj), shown in the REPL
__len__     len(obj)
__add__     obj + other
__eq__      obj == other
```

```python
class Vec:
    def __init__(self, x):  self.x = x
    def __str__(self):  return f'Vec({self.x})'
    def __eq__(self, o):  return self.x == o.x

print(Vec(3))        # Vec(3)
Vec(3) == Vec(3)     # True
```

8 Protected and private attributes

```
class MyClass:
    def __init__(self, value):
        self._protected = value    # internal use
        self.__private  = value    # name-mangled
    def get_private(self):
        return self.__private

obj = MyClass(10)
obj._protected             # 10  ('hands off')
obj._MyClass__private      # name-mangled access
obj.get_private()          # 10  (preferred)
```

- Single underscore `_` is a convention meaning “internal”
- Double underscore `__` mangles `__x` to `_ClassName__x`
- Neither is truly hidden; Python enforces nothing

9 Properties

Properties give attribute-style access while running code:

```
class Dog:
    def __init__(self, age):
        self._age = age
    @property
    def age(self):          # getter
        return self._age
    @age.setter
    def age(self, value):       # setter
        if value < 0:
            raise ValueError('age must be >= 0')
        self._age = value
```

- `d.age` reads through the getter
- `d.age = 5` runs the setter, which validates
- `@age.deleter` likewise handles `del d.age`

10 Data classes

```python
from dataclasses import dataclass

@dataclass
class Point:
    x: float
    y: float

p = Point(1.0, 2.0)
str(p)                  # 'Point(x=1.0, y=2.0)'
p == Point(1.0, 2.0)    # True
```

- `@dataclass` writes `__init__`, `__repr__`, `__eq__`
- Fields can have defaults, e.g. `kind: str = 'dog'`
- Defaulted fields must come after the others

11 Inheritance

```
class Animal:
    def __init__(self, name):
        self.name = name

class Dog(Animal):              # inherits __init__
    def speak(self):
        return f'{self.name} woofs'

class Puppy(Dog):
    def __init__(self, name, owner):
        super().__init__(name)    # run parent init
        self.owner = owner
```

- A subclass inherits the parent's attributes and methods
- Override a method by redefining it in the subclass
- `super()` calls the parent's version of a method

12 Composition

A class can *contain* another object instead of inheriting:

```
class Engine:
    def start(self):
        return 'vroom'

class Car:
    def __init__(self):
        self.engine = Engine()    # composition
    def drive(self):
        return self.engine.start()

car = Car()
car.drive()     # 'vroom'
```

- Prefer composition when the relationship is “has a”, not “is a”
- A `Car` *has an* `Engine`; it is not an `Engine`

Further reading: `__repr__` vs `__str__`

```
class Dog:
    def __init__(self, name, age):
        self.name = name
        self.age  = age
    def __repr__(self):
        return f'Dog({self.name!r}, {self.age!r})'
    def __str__(self):
        return f'{self.name} (age {self.age})'

d = Dog('Rex', 5)
repr(d)    # dev-facing, unambiguous
str(d)     # user-facing, readable
```

- `__repr__` is shown in the REPL; prefer it if you define only one
- `print()` uses `__str__`, falling back to `__repr__` if absent

Further reading: frozen data classes and `field()`

```python
from dataclasses import dataclass, field

@dataclass(frozen=True)
class Point:
    x: float
    y: float

p = Point(1.0, 2.0)
p.x = 3.0   # error: instance is immutable

@dataclass
class Kennel:
    dogs: list = field(default_factory=list)
```

- `frozen=True` makes instances immutable and hashable
- `field(default_factory=list)` avoids the mutable-default trap from lesson 6